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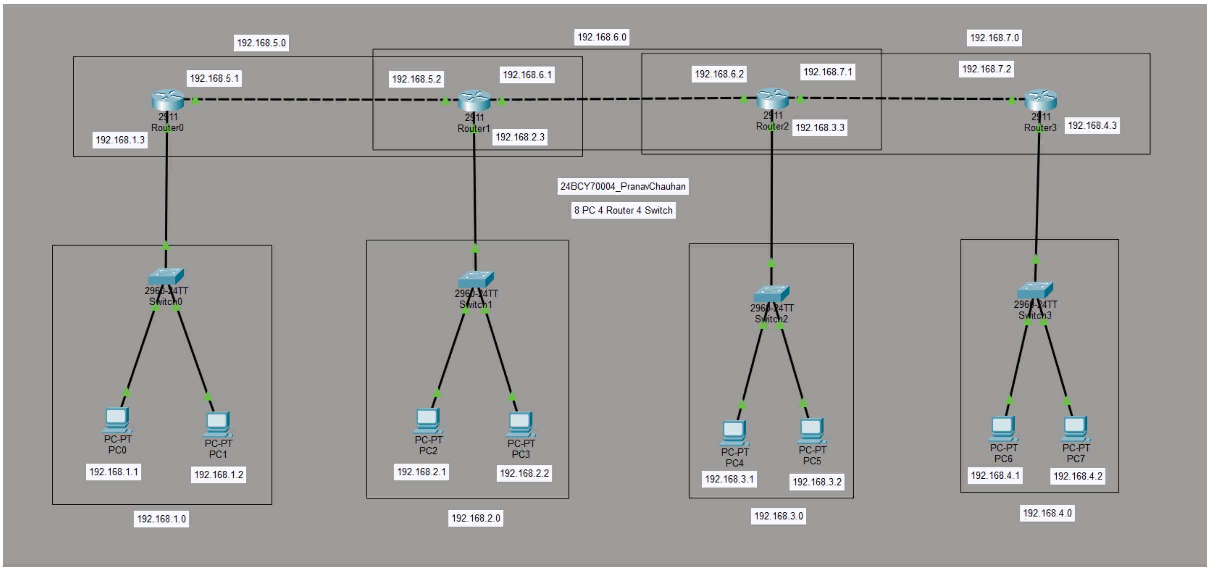
Question 1: Develop Static Routing with 4 PC's, 4 Switches & 8 PC's.

Make sure to highlight the Networks with their network ID's.



1. Network Addressing

Network	Network ID	Devices
LAN 1	**192.168.1.0/24**	PC0, PC1, Router0
LAN 2	**192.168.2.0/24**	PC2, PC3, Router1
LAN 3	**192.168.3.0/24**	PC4, PC5, Router2
LAN 4	**192.168.4.0/24**	PC6, PC7, Router3
R0–R1	**192.168.5.0/24**	Router0, Router1
R1–R2	**192.168.6.0/24**	Router1, Router2
R2–R3	**192.168.7.0/24**	Router2, Router3

2. IP Addresses

PC Network 1 – 192.168.1.0/24

- PC0 → 192.168.1.1
- PC1 → 192.168.1.2
- Router0 LAN → 192.168.1.3
- Default Gateway → 192.168.1.3

PC Network 2 – 192.168.2.0/24

- PC2 → 192.168.2.1
- PC3 → 192.168.2.2
- Router1 LAN → 192.168.2.3
- Default Gateway → 192.168.2.3

PC Network 3 – 192.168.3.0/24

- PC4 → 192.168.3.1
- PC5 → 192.168.3.2
- Router2 LAN → 192.168.3.3
- Default Gateway → 192.168.3.3

PC Network 4 – 192.168.4.0/24

- PC6 → 192.168.4.1
- PC7 → 192.168.4.2
- Router3 LAN → 192.168.4.3
- Default Gateway → 192.168.4.3

Router-to-Router IPs

- **R0 ↔ R1:** 192.168.5.1 ↔ 192.168.5.2
- **R1 ↔ R2:** 192.168.6.1 ↔ 192.168.6.2
- **R2 ↔ R3:** 192.168.7.1 ↔ 192.168.7.2

Subnet mask for all networks:

255.255.255.0

Question 2: Write the difference between Hub, Switch & Router in Tabular Format.Explain why router is the most intelligent & Hub is least intelligent device. (3 M)

Answer:

Feature	**Hub**	**Switch**	**Router**
OSI Layer	Physical Layer (Layer 1)	Data Link Layer (Layer 2)	Network Layer (Layer 3)
Address Used	No address	MAC Address	IP Address
Data Transmission	Broadcasts data to all ports	Sends data only to the destination port	Routes packets between different networks
Intelligence	Least intelligent	More intelligent than Hub	Most intelligent
Collision	High chance of collision	Reduces collisions	Separates networks and prevents broadcast between them
Network Connection	Connects devices in a LAN	Connects devices in a LAN	Connects different networks/LANs
Traffic Handling	Poor	Better	Best
Example	Old/simple LAN device	Ethernet LAN switch	Connects LAN to another LAN/Internet

Why is Hub least intelligent?

A **Hub** simply receives data and broadcasts it to **all connected devices** without checking the destination. It does not use MAC or IP addresses, so it has very little decision-making capability.

Why is Router most intelligent?

A **Router** examines the **destination IP address** of packets and uses its routing table to determine the best path to another network. It can connect different networks and make routing decisions, so it is considered the **most intelligent** among the three.

In short:

Hub → Broadcasts to everyone → Least intelligent
Switch → Uses MAC address → Moderately intelligent
Router → Uses IP address + routing table → Most intelligent

Question 3: What are Random Access Protocols? Why do we need them? Explain each category in detail with diagram and example. Also Explain difference between CSMA/CD & CSMA/CA. (3 M)

Answer:

What are Random Access Protocols?

Random Access Protocols are MAC (Medium Access Control) protocols used when multiple devices share the same communication channel. Any device can access the channel when it has data to send, but a mechanism is required to handle situations where **two or more devices transmit at the same time**.

Why do we need them?

We need random access protocols to:

- Control access to a shared communication medium.
- Reduce or handle **collisions**.
- Improve network efficiency.
- Allow multiple devices to communicate over the same channel.

2. Categories of Random Access Protocols

Random Access Protocols are mainly divided into:

```
                 Random Access Protocols
                           |
             ┌─────────────┼─────────────┐
             ↓             ↓             ↓
           ALOHA          CSMA        CSMA/CD
             |             |
       ┌─────┴─────┐       |
       ↓           ↓       └── CSMA/CA
  Pure ALOHA   Slotted
               ALOHA
```

A. ALOHA

ALOHA allows a device to **transmit whenever it has data**.

Pure ALOHA

```
PC1 ───────┐
PC2 ───────┼─── Shared Channel ── Receiver
PC3 ───────┘
             ↑
        Collision may occur
```

- A device transmits whenever it wants.
- If two devices transmit simultaneously, a **collision** occurs.
- After collision, the devices wait for a random time and retransmit.

Example: Early satellite communication systems.

Slotted ALOHA

The channel is divided into fixed **time slots**.

```
Time →      | Slot 1 | Slot 2 | Slot 3 | Slot 4 |
PC1 ───────|   TX    |        |        |        |
PC2 ───────|         |   TX   |        |        |
PC3 ───────|         |        |   TX   |        |
```

A device can transmit **only at the beginning of a time slot**, reducing collisions compared with Pure ALOHA.

B. CSMA

CSMA = Carrier Sense Multiple Access

```
     Is channel free?
            |
      ┌─────┴─────┐
     YES          NO
      ↓            ↓
   Transmit       Wait
```

Before transmitting, a device **listens to the channel**.

If the channel is free → transmit.
If the channel is busy → wait.

Example: Ethernet networks use CSMA-based access mechanisms.

CSMA has different approaches such as **1-persistent, non-persistent and p-persistent CSMA**.

C. CSMA/CD

CSMA/CD = Carrier Sense Multiple Access with Collision Detection

It is used in traditional **shared, half-duplex Ethernet**.

```
Listen → Channel free → Transmit
                    ↓
             Detect collision?
                ↙       ↘
              NO          YES
              ↓            ↓
          Continue      Stop + Wait
                           ↓
                       Retransmit
```

Working:

1. Device listens to the channel.
2. If the channel is free, it transmits.
3. While transmitting, it checks for a collision.
4. If collision occurs, transmission stops.
5. Device waits for a random backoff time and retransmits.

D. CSMA/CA

CSMA/CA = Carrier Sense Multiple Access with Collision Avoidance

It is mainly used in **Wi-Fi (IEEE 802.11)**.

Unlike CSMA/CD, it tries to **avoid collisions before they happen**, because detecting collisions while transmitting wirelessly is difficult.

```
Listen
   ↓
Channel free?
   ↓
Wait / Random Backoff
   ↓
Transmit
   ↓
ACK received?
  ↙       ↘
 YES       NO
  ↓         ↓
Success   Retransmit
```

3. CSMA/CD vs CSMA/CA

Feature	**CSMA/CD**	**CSMA/CA**
Full form	Carrier Sense Multiple Access with Collision Detection	Carrier Sense Multiple Access with Collision Avoidance
Mainly used in	Traditional Ethernet	Wi-Fi
Main idea	**Detect** collision	**Avoid** collision
Collision handling	Detects collision after it occurs	Uses waiting/backoff and mechanisms such as ACK to reduce collisions
Transmission medium	Wired	Wireless
Acknowledgement	Generally not required for basic Ethernet delivery	ACK is commonly used in Wi-Fi
Example	Shared half-duplex Ethernet	IEEE 802.11 Wi-Fi